

Final Student Evaluation Workflow

1. Workflow Overview

The “Final Student Evaluation Workflow” is a composite service in the School Management System that combines multiple independent services to evaluate a student’s final academic status.

The workflow integrates:

- Attendance & Presence Service
- Grade Service

The final result is a unified Completion Report that contains:

- Attendance eligibility
- Academic grades
- Final completion decision

2. Participating Services

Service	Responsibility
Attendance Service	Checks attendance eligibility
Grade Service	Retrieves academic grades
Final Evaluation Workflow Combines results into one final report	

3. Step-by-Step Workflow Logic

Step 1 — Attendance Verification

The workflow starts by calling the Attendance Service.

Process

- The system receives a Student ID.
- The Attendance Service calculates attendance percentage.
- The service checks whether the student satisfies the required attendance threshold.

Example Rule

Minimum Attendance Requirement = 75%

Output

- Eligible
 - Not Eligible
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Step 2 — Grade Retrieval

If the student is eligible:

- The workflow calls the Grade Service.

Process

The Grade Service retrieves:

- Assignment grades
- Midterm grades
- Final exam grades

Output

A complete academic transcript for the semester.

Step 3 — Completion Report Generation

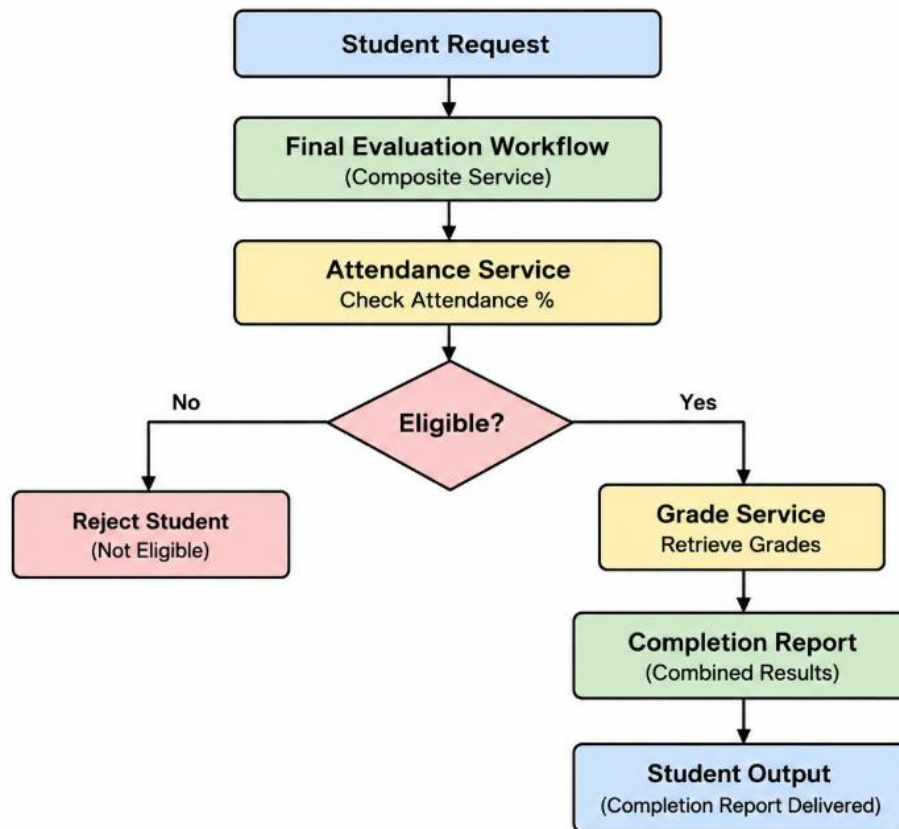
The workflow combines:

- Attendance result
- Academic grades
- Final evaluation status

Final Output

Completion Report

4. Service Composition Diagram



5. Architectural Justification

This workflow follows Service-Oriented Architecture principles because:

- Each service is independent and reusable
- Services communicate through clear interfaces
- The workflow supports scalability and maintainability
- Additional services can be integrated easily in the future

6. Conclusion

The “Final Student Evaluation Workflow” demonstrates how multiple services collaborate to automate the academic evaluation process efficiently. By combining attendance verification and grade retrieval, the system generates accurate completion reports while maintaining modularity, scalability, and maintainability.